RYAN CHO

cho6@sas.upenn.edu | Personal website

EDUCATION

University of Pennsylvania | Expected graduation 2029 | GPA 4.0/4.0

Double major in Biology and Computer Science, Chemistry minor

Illinois Math and Science Academy | GPA 3.94/4.0

AWARD HIGHLIGHTS

Grant Winner: Wharton Venture Innovation Fund. IVision: U-Net deep learning for vein visualization in clinical settings.

First Place: National Junior Science and Humanities Symposium. Math and Computer Science category, from 8,000+ initial competitors. Applied natural language processing, signal processing, and machine learning to data from 38 epilepsy patients to study clinical applications of deep brain stimulation.

Top Score: SIMIODE International Challenge Using Differential Equations Modeling. Problem C, approximately 500 undergraduate and high school competitors. Modeled dog behavior using calculus-based physics.

RESEARCH HIGHLIGHTS

First Author Peer Reviewed Publication. "Differentiating hand gestures from forearm muscle activity using machine learning." *International Journal of Occupational Safety and Ergonomics*, Taylor & Francis, Aug 2024. doi: 10.1080/10803548.2024.2383021

First Author Peer Reviewed Publication. "Investigating brain activity patterns during learning tasks through EEG and machine learning analysis." *International Journal of Information Technology*, Springer, Apr 2024. doi: 10.1007/s41870-024-01856-4

Patent Co Inventor. "Visual assistance system." Developed microcontroller and sensor components to assist the deaf and blind community. Sep 2026. Patent information

First Author Conference Proceedings and Oral Presentation. "Neural Understanding of STEM Activities." *IEEE International Symposium on Computer Science and Educational Technology*, Laubusch, Germany, Jul 2024. doi: 10.1109/ISCSET58624.2024.10807951

EMPLOYMENT

Penn Center for Neuroengineering and Therapeutics Dec 2025 – Present

Research Assistant

- Received a Penn CURF Summer Research Grant in 2026.
- Created a bedside clinical risk calculator that processes clinical and EEG information entered by neurologists. Developed and statistically evaluated predictive algorithms using data from 1,300 epilepsy patients, achieving an AUROC of approximately 0.80. EEG analysis code | Clinical calculator code
- Built an easy-to-use graphical interface for clinicians to view and annotate EEGs, supporting evaluation of an interictal spike localization method in a study of 200 epilepsy patients. GUI code
- Analyze lateralized periodic discharges along the ictal-interictal continuum and machine learning algorithms for identifying where seizures begin in the brain.

International Youth Neuroscience Association Oct 2023 – May 2025

Website Lead

- Collaborated with team members on website design and layout. Independently rebuilt the Wix website as a custom-coded site serving 1,300 unique annual visitors and saving approximately $300 annually. Created Lottie animations and design templates. youthneuro.org